

DeFi — Decentralized Finance

Introduction, Architecture, Examples and Key Concepts

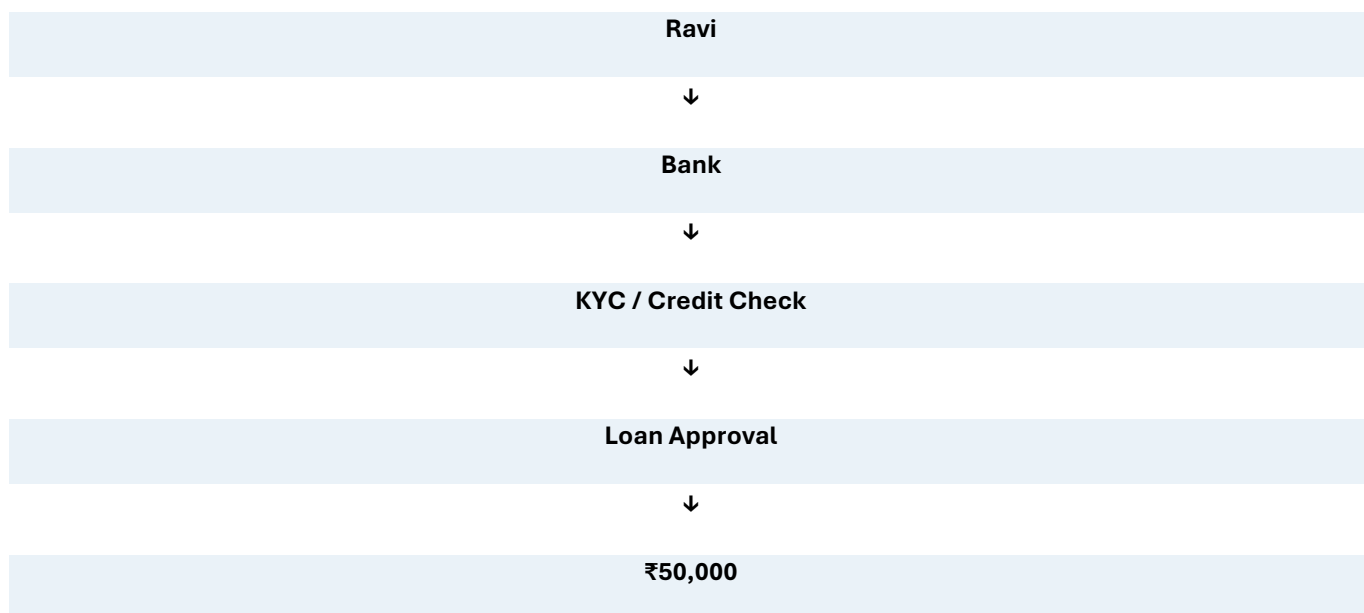
1. What is DeFi?

DeFi (Decentralized Finance) refers to financial applications built primarily on blockchain networks that allow people to perform financial activities such as lending, borrowing, trading, exchanging assets, and earning returns without relying on traditional intermediaries such as banks or brokers.

2. Simple Example

Suppose Ravi wants to borrow ₹50,000.

In traditional finance:



The bank acts as the intermediary.

In DeFi, the conceptual flow is:





Blockchain



Liquidity Pool



Crypto Loan

Instead of a bank employee approving and processing the transaction, smart contracts execute predefined rules on the blockchain.

An important distinction for students is that DeFi loans are usually crypto-asset loans denominated in tokens/stablecoins, not conventional unsecured rupee bank loans. Many DeFi lending systems require the borrower to provide crypto collateral.

3. Main Components of DeFi

USER



Wallet — MetaMask etc.



DeFi DApp — Web / App



Smart Contracts



Blockchain — Ethereum etc.

The wallet manages the user's blockchain account and signs transactions. The DApp provides the user interface. Smart contracts implement the financial rules, while the blockchain records and executes transactions.

4. What Can We Do Using DeFi?

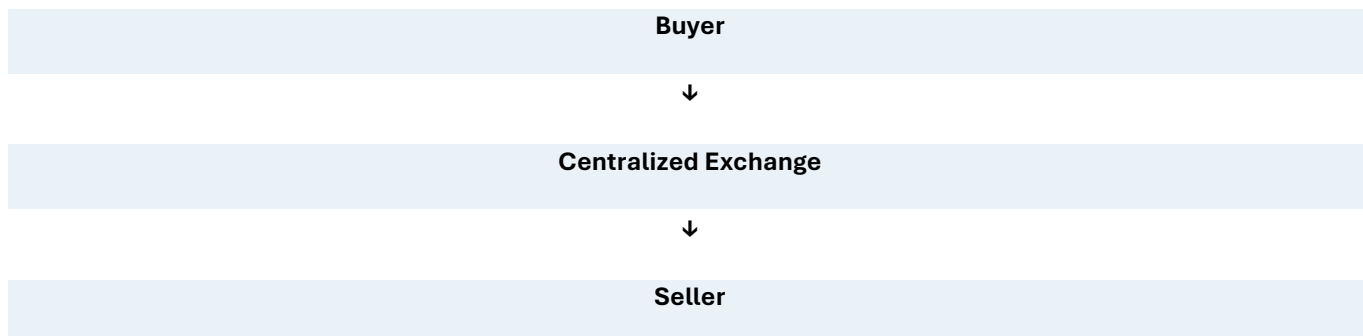
DeFi Service	Purpose	Example
Decentralized Exchange	Exchange crypto assets	Uniswap
Lending	Supply assets and earn interest	Aave
Borrowing	Borrow assets using collateral	Aave
Stablecoins	Blockchain assets designed for price stability	DAI, USDC
Liquidity Provision	Supply tokens to trading pools	Uniswap
Staking	Participate in blockchain/economic security mechanisms	Ethereum staking
Yield strategies	Seek returns using DeFi protocols	Various DeFi protocols

Derivatives	Trade blockchain-based derivative products	dYdX and others
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5. Example: Decentralized Exchange

One well-known DeFi application is Uniswap (<https://uniswap.org/>).

In a traditional exchange model:



A decentralized exchange can instead use smart contracts and liquidity pools:



For example, a user might exchange:

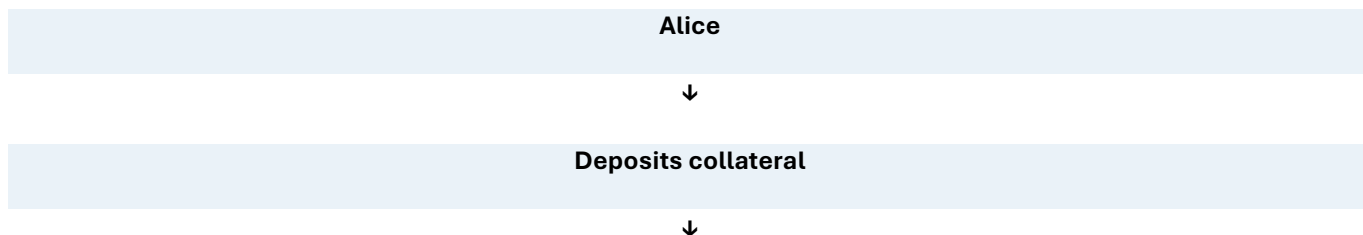
ETH → USDC

The transaction is executed through smart contracts rather than a traditional broker.

6. Example: DeFi Lending

Aave (<https://aave.com/>) is a well-known decentralized lending protocol.

Imagine Alice has crypto assets but needs liquidity.



Aave Smart Contract



Collateral Pool



Borrow USDC / other asset

Alice may deposit an accepted crypto asset as collateral and borrow another asset against it.

For example:

Deposit ETH



ETH becomes collateral



Borrow USDC



Pay borrowing interest



Repay USDC



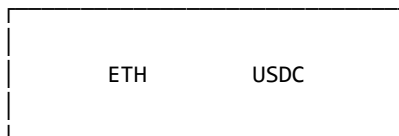
Withdraw collateral

The exact collateral requirements, rates, and liquidation rules depend on the protocol and market.

7. What is a Liquidity Pool?

Liquidity pools are one of the most important concepts in DeFi.

ETH / USDC Pool



↑
Liquidity Providers

Users deposit assets into the pool. Other users can then trade against that liquidity.

Liquidity providers may receive a share of trading fees or other incentives, but they also face risks such as impermanent loss, token price movements, and smart-contract vulnerabilities.

8. Stablecoins in DeFi

Because cryptocurrencies such as ETH can fluctuate significantly, DeFi commonly uses stablecoins.

A stablecoin attempts to maintain a relatively stable value, often relative to a currency such as the US dollar.

Example — USDC:

1 USDC $\approx$ 1 USD

DAI is another important example. It is associated with the Maker ecosystem and uses blockchain-based mechanisms designed to maintain its peg.

Stablecoins are widely used for lending, borrowing, trading and payments within DeFi.

9. Traditional Finance vs DeFi

Traditional Finance	DeFi
Banks and financial institutions	Blockchain protocols
Bank account	Crypto wallet
Centralized control	Often decentralized or distributed governance
Institution maintains records	Blockchain records transactions
Bank processes transactions	Smart contracts execute transactions
Usually permissioned access	Many protocols are permissionless
Limited banking/business hours for some operations	Blockchain protocols generally operate 24×7
Rules implemented by institutions/software	Rules heavily implemented through smart contracts

DeFi should not be interpreted as meaning “no rules” or “no risk.” DeFi applications can still have governance structures, administrators, regulatory considerations, technical dependencies, and significant financial risks.

10. Why Ethereum is Important to DeFi

Ethereum is one of the major ecosystems for DeFi because it supports programmable smart contracts.

A simple Solidity contract demonstrates the underlying idea:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.28;

contract SimpleBank {

    mapping(address => uint256) public balances;

    function deposit() public payable {
        balances[msg.sender] += msg.value;
    }

    function withdraw(uint256 amount) public {
        require(balances[msg.sender] >= amount,
            "Insufficient balance");

        balances[msg.sender] -= amount;
    }
}
```

```

    payable(msg.sender).transfer(amount);
  }
}

```

This is not a production DeFi protocol, but it demonstrates the fundamental concept: financial logic can be encoded in a smart contract.

11. Popular DeFi Examples

Protocol	Website	Main Use
Uniswap	https://uniswap.org/	Decentralized token exchange and liquidity pools.
Aave	https://aave.com/	Decentralized lending and borrowing.
Maker	https://makerdao.com/	Ecosystem associated with the DAI stablecoin.
Lido	https://lido.fi/	Liquid staking protocol.

These protocols show that DeFi is not a single application. It is an ecosystem of financial smart contracts and applications.

12. A Useful Definition for Students

“DeFi is a blockchain-based financial ecosystem in which smart contracts provide services such as trading, lending, borrowing and asset management, reducing or replacing the role of traditional financial intermediaries.”

13. Suggested Practical Exercise

A useful next practical exercise is to build a Mini DeFi Bank using Solidity and Hardhat 3 with `deposit()`, `withdraw()`, `getBalance()`, lending, borrowing and interest calculation. This connects DeFi concepts directly with Solidity smart contracts and the Hardhat development workflow.